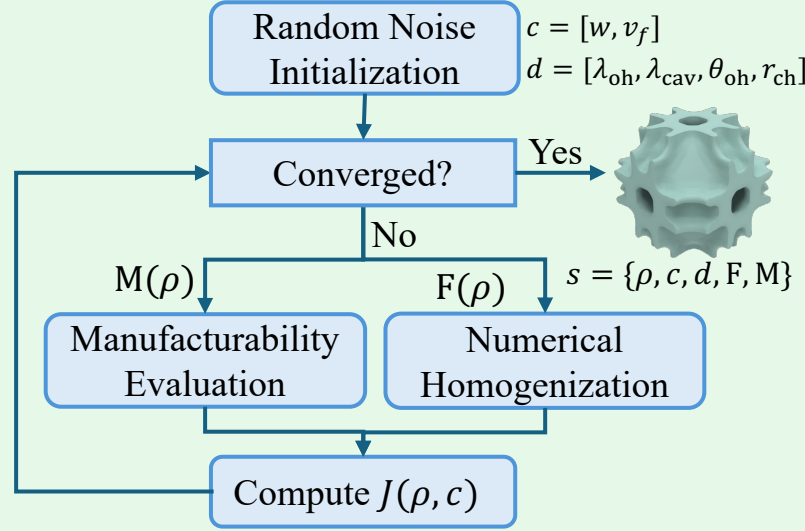
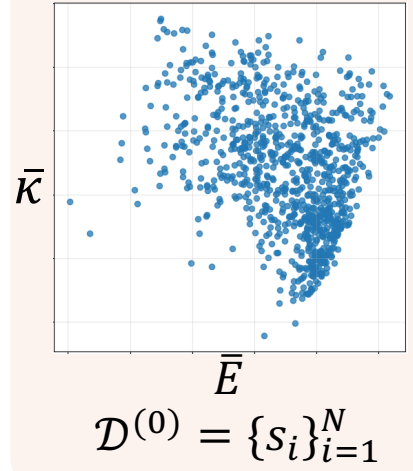


Stage I: Random Exploration and Initial Dataset Construction

Lattice Optimization

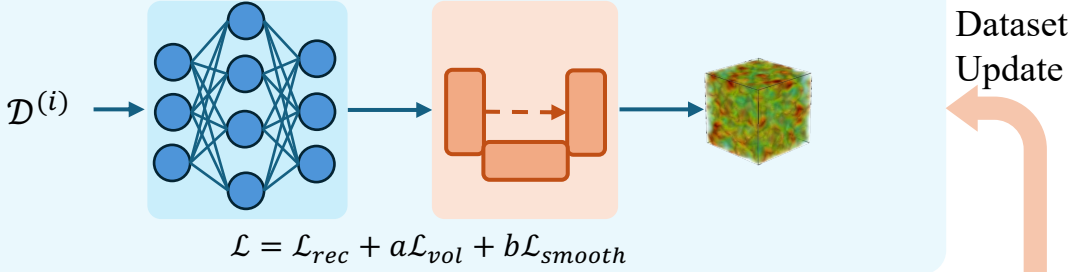


Initial Dataset Construction

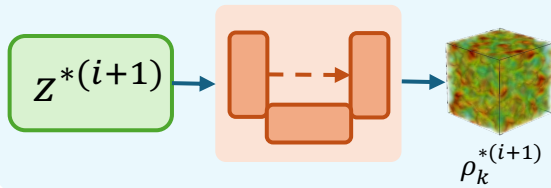


Stage II: Network-Guided Iterative Pareto-Front Construction

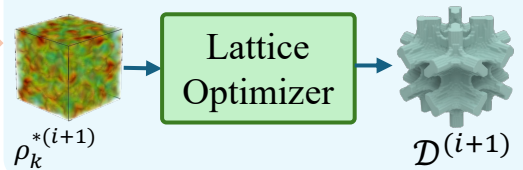
Dataset Update and Network Retraining (or Initial Training with $\mathcal{D}^{(0)}$)



Latent-Space Decoding and Initial Density Field Generation



Topology Optimization and Pareto-Front Update



Pareto-Front Region Identification and Adaptive Interpolation

